CSC 316

Handout: The Design Critique Checklist (Growing)

What this is. One checklist you run over *every* visualization you make — yours or someone else's. It **grows as the course does**: each lab switches on a new section, and by the end you'll hold a full design-critique checklist. **Companion to every lab.** Keep it open when you reach the **Judge** step; that's what it's for.

> **How to read it.** Each section header names the lab **and the date** it switches on. Only the sections at or before your current lab are **live** — run those, and *skip the rest*: they use vocabulary the course hasn't taught yet, and they'll make sense when their lab arrives. Once a section is on, it stays on — a Lab 5 story is judged against §1 through §5, and by Lab 7 you run the whole thing. All seven sections are printed here (this is the only copy you'll ever need — nothing to re-download) so you can see what's coming as well as what's already live.

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§1 · Truthful & readable — *from Lab 1 (Fri Sep 18)*

The floor every chart must clear. If any of these fail, nothing else matters.

- ❍ **Answers the question.** A stranger could state the takeaway in about ten seconds.
- ❍ **Truthful to the data.** No silently dropped rows; no invented numbers; the *right* "time" column; any counting / cleaning choice is stated on the chart; nothing mislabeled (e.g. a duration masquerading as a rating).
- ❍ **Readable.** Title; axis labels **with units**; a legend; sensible sorting; nothing important hidden by overlap.
- ❍ **Right chart for the question.** Trend → line; ranking → sorted bars; composition → stacked bar; relationship → scatter.
- ❍ **It runs.** Opens / builds with no errors; a grader can actually see it.

§2 · Comparing sketches — *from Lab 2 (Fri Sep 25)*

Use these checks for the two static code sketches.

- ❍ **Design matches the intent.** The marks, visual channels, and layout follow your drawings or a change you deliberately chose.
- ❍ **Separate components, same data.** Sketch A and Sketch B receive the same rows and use the same measure and comparison periods.
- ❍ **The question can be answered.** Test the data question using each sketch. Note which comparisons are easier or harder in each.
- ❍ **Numbers are checked.** Compare plotted values with the supplied data and known answers.
- ❍ **Runs clean.** `npm run dev` with no console errors.

§3 · Encoding & channels — *from Lab 3 (Fri Oct 2)*

The mapping from data columns to things you can see.

- ❍ **Every channel names a column.** For each of x, y, size, color, shape: which column drives it, or is it a constant?
- ❍ **Scale type fits the data type.** Quantitative → linear / sqrt / log / sequential; categorical → ordinal.
- ❍ **Size is read as area.** Use `scaleSqrt`; anchor domain **and** range at zero if you want area *proportional* to value; add a legend when the quantity is logarithmic.
- ❍ **Position points the right way.** Mind the SVG y-flip (range `[height, 0]`); nothing clipped at the edges.
- ❍ **Strongest channels carry the most important variables.** Position first; color / shape last.

- ❍ **Legible mapping back.** Legends and axis units let a reader decode every channel.
- ❍ **No lying channels.** Length has a zero baseline; size uses area; color's long tail doesn't wash out the middle.
- ❍ **You can point to the code.** The `{#each}` that binds a row to a mark, and a `scale(d.field)` call that turns a value into a pixel.

(Whether the palette is the right kind *— and whether it survives a color-vision check — is §6 and §7. Here, color only has to name a column and not lie about magnitude.)*

§4 · Interaction & coordination — *from Lab 4 (Fri Oct 9)*

Once several views share one selection and answer to the user.

- ❍ **Named components.** Each has a clear purpose; repeated UI uses one reusable component.
- ❍ **Truthful interaction.** Filtering preserves the data and its meaning; absent data stays absent, and aggregation choices are stated.
- ❍ **There is exactly one selection state.** You can point to the single `$state` every view depends on.
- ❍ **Every view derives from it.** No view reads the global data while a sibling is filtered (the silent frozen-view lie).
- ❍ **Link vs. filter is intentional.** Each interaction is *deliberately* a highlight (dim the rest) or a removal (recompute) — not by accident.
- ❍ **Filtered aggregates recompute.** KPIs and bar heights reflect the subset, not the whole.
- ❍ **The selection can be cleared.** There's an obvious way back to the full view (clear the brush, a reset button).
- ❍ **Interaction is discoverable.** A first-time user can tell what's clickable / brushable (cursor, hint, hover feedback); nothing traps them.
- ❍ **No regressions.** The interaction that worked last commit still works — and there's a green commit to prove it.
- ❍ **You can point to the code.** The `$state` that holds the selection, and a `$derived` that reads it.

§5 · Story & annotation — *from Lab 5 (Fri Oct 23)*

Once the piece leads a reader through a point (scrollytelling and beyond).

- ❍ **One idea per step / scene.** No step asks the reader to absorb two movements at once.
- ❍ **The graphic is driven by one state.** Scroll (or the control) writes one `step` /selection; the graphic is `$derived` from it, never updated imperatively.
- ❍ **The graphic is sticky** and holds through each step, rather than scrolling away from the text that describes it.
- ❍ **Every text beat is a checked claim.** You can name the column and year behind each sentence.
- ❍ **No invented trend.** No "every / always / kept" the line doesn't support; no zoom sold as the whole truth.
- ❍ **No invented cause.** No "because X" unless a column contains X.
- ❍ **Reframings are labeled.** The reader always knows which measure a scene shows (e.g. count vs. rate).
- ❍ **The title says the finding, not the topic**; annotations direct attention to the point.

§6 · Perception & critique — *from Lab 6 (Fri Nov 6)*

Once you judge the *rendered image*, not the code — the things your collaborator can't reliably call.

- ❍ **Pop-out.** If something should stand out, does it — instantly? Is it one strong channel against neutral distractors, not a conjunction of two?
- ❍ **Channel budget.** How many channels carry how many variables? Is anything encoded redundantly by accident (clutter) — or *not* redundantly where robustness matters?
- ❍ **Effectiveness.** Is the most important comparison on **position**? Is any key magnitude stranded on area or color?
- ❍ **Categorical vs. ordered.** Do unordered categories use distinct hues rather than a ramp? Do ordered values use an ordered channel?

- ❍ **Colorblind-safe.** Checked under **deuteranopia** and **protanopia** — are all categories still distinguishable?
- ❍ **Not color alone.** Is category also carried by shape, label, or position, so it survives greyscale?
- ❍ **Contrast.** Is text ≥ 4.5:1? Are marks clearly figure against ground?
- ❍ **You beat the agent's self-grade.** Did you find at least one perceptual failure it rated as fine — and which *tool* proved it?

§7 · Color & colormaps — *from Lab 7 (Fri Nov 13)*

The last section: not whether the colors are *safe*, but whether they're the *right kind*.

- ❍ **Family matches type.** Nominal → categorical; ordinal or quantitative → sequential; signed with a meaningful zero → diverging. Can you name the type from the data, not from the look?
- ❍ **Ordered variables are on an ordered scale.** No ranked variable is wearing a categorical palette; the rank order used is the *real* one, not alphabetical.
- ❍ **Diverging maps are anchored.** The neutral color sits exactly on the meaningful value, and the domain is symmetric about it.
- ❍ **No rainbow.** No `jet`-style map on continuous data; equal steps in the data look like equal steps.
- ❍ **Category count is within capacity.** No more than ~8 hues carrying nominal categories; beyond that, group, facet, or change channel.
- ❍ **Skew handled in the domain, not the palette.** A right-skewed quantity is transformed (log/sqrt) rather than left to bunch at one end of the ramp.
- ❍ **The legend is decodable.** Ordered by rank; ends labeled in words; units named; the zero marked on any diverging scale; text ≥ 4.5:1.
- ❍ **Color survives the checks.** Deuteranopia, protanopia, greyscale — and color is never the only signal. *(§6's tools, run on §7's palette.)*
- ❍ **Right encoding, right quantity.** The variable being colored means what the chart implies — no confound doing the talking.

> **Using it in a lab.** At the **Judge** step, walk every *active* section top to bottom. Turn each failing box into a precise next frame ("*the size legend is missing; add M6/M7/M8 reference circles*"). Then — the course habit — ask **Claude to run the same checklist on its own output** before you do, and invite a **tablemate** to spot-check it too. Peer feedback is optional in Lab 2; from Lab 3, record it in process-book Q7. The checklist gives you, Claude, and your tablemate the same criteria.

> **What this checklist can't tell you.** Every box above is something *you* can judge by looking. None of them tells you whether a reader who has never seen the chart gets the finding out of it — a chart can pass all seven sections and still hand somebody a false belief. That's the other instrument: see Think-Aloud & Comprehension Testing, and Lab 8, where you point a study at this checklist and find its ceiling.